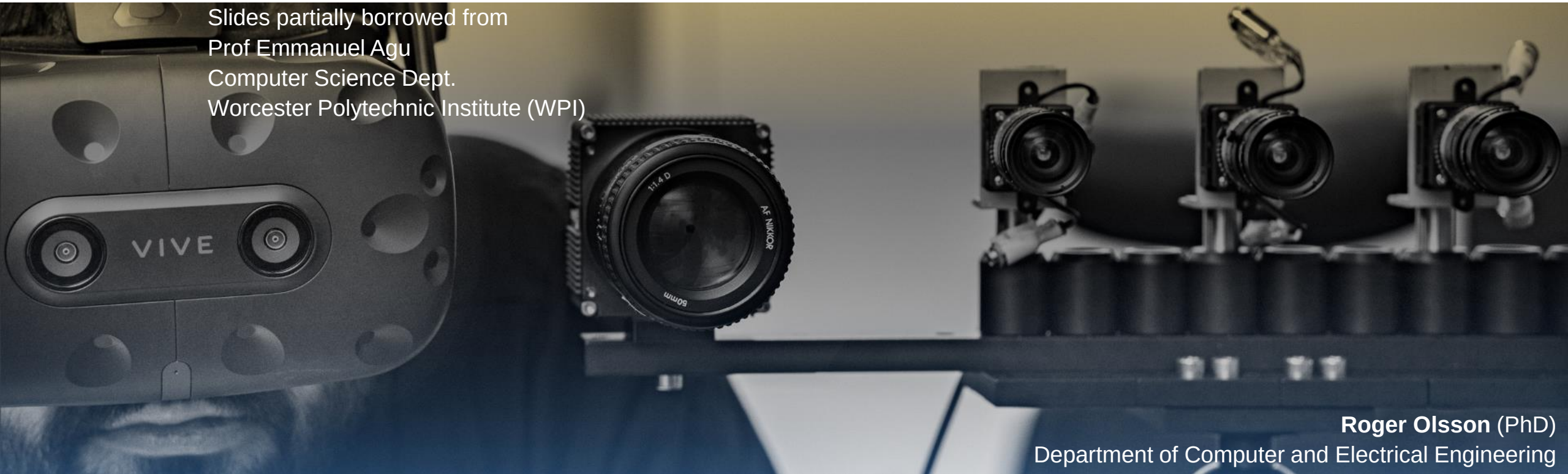
Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Lecture 11 - Spatial domain image processing IV



Slides partially borrowed from
Prof Emmanuel Agu
Computer Science Dept.
Worcester Polytechnic Institute (WPI)

Roger Olsson (PhD)
Department of Computer and Electrical Engineering

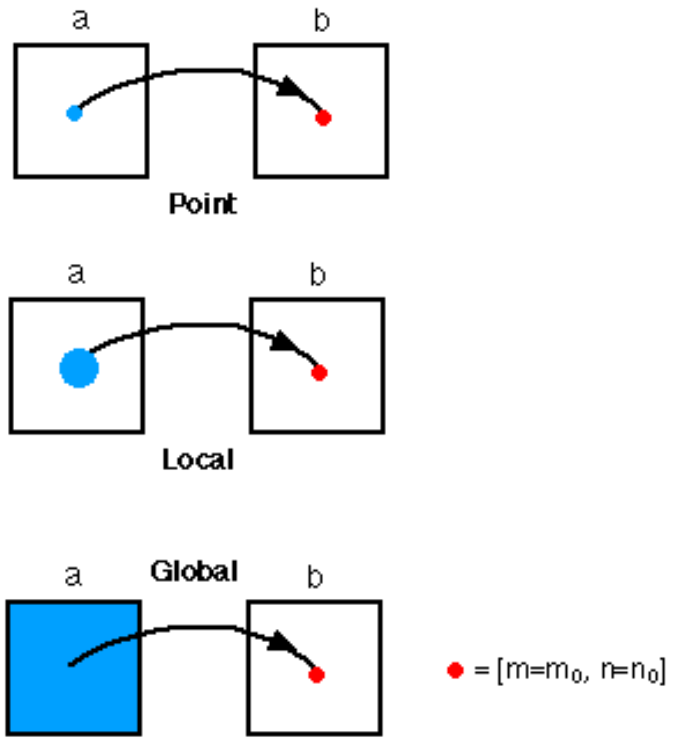
Image processing revisited

Types of image enhancement operations

Point/pixel operations
Output value at specific pixel (x,y) is dependent only on the input value at (x,y)

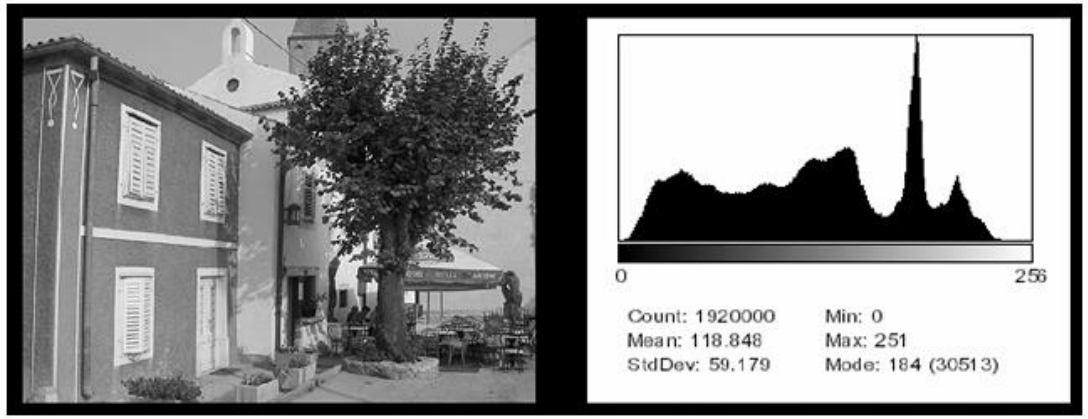
Local operations
The output value at pixel (x,y) is dependent on the input values in the *neighborhood* of (x,y)

Global operations
The output value at (x,y) is dependent on all the values in the input image

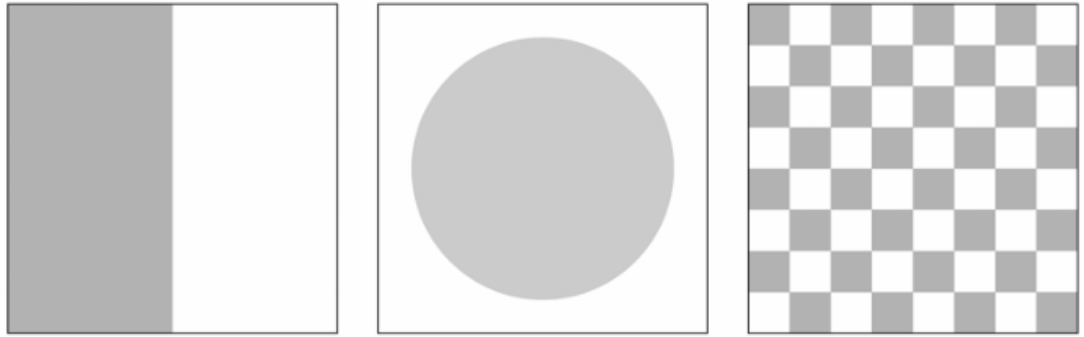


Histogram

- Analogous to Probability Density Function (PDF)
- How many times (frequency) does each intensity value appear in the image



- Shows valuable information about:
  - Over and under exposure
  - Brightness
  - Contrast
  - Dynamic range
  - Hints about processing performed
- Shows no indikation of location of pixels

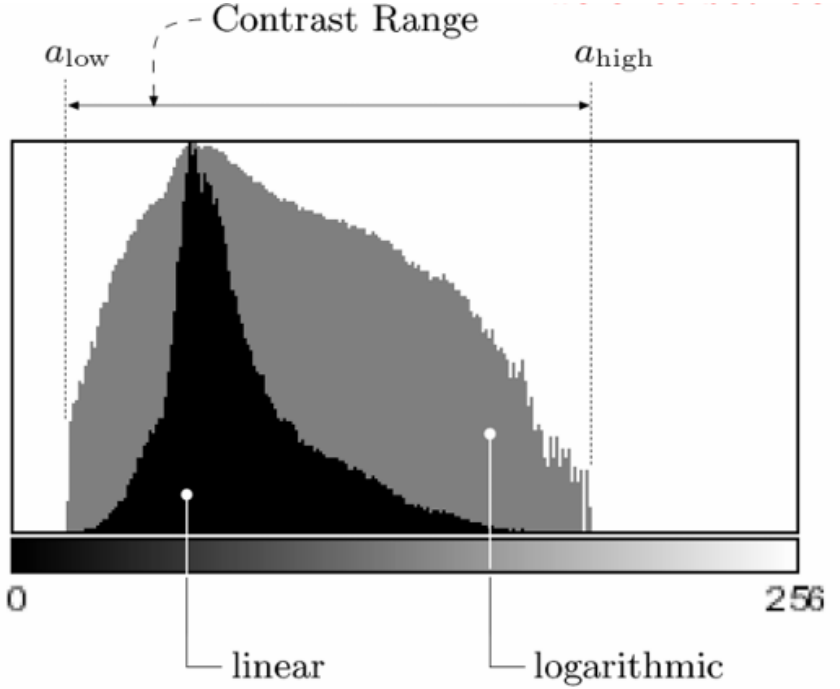


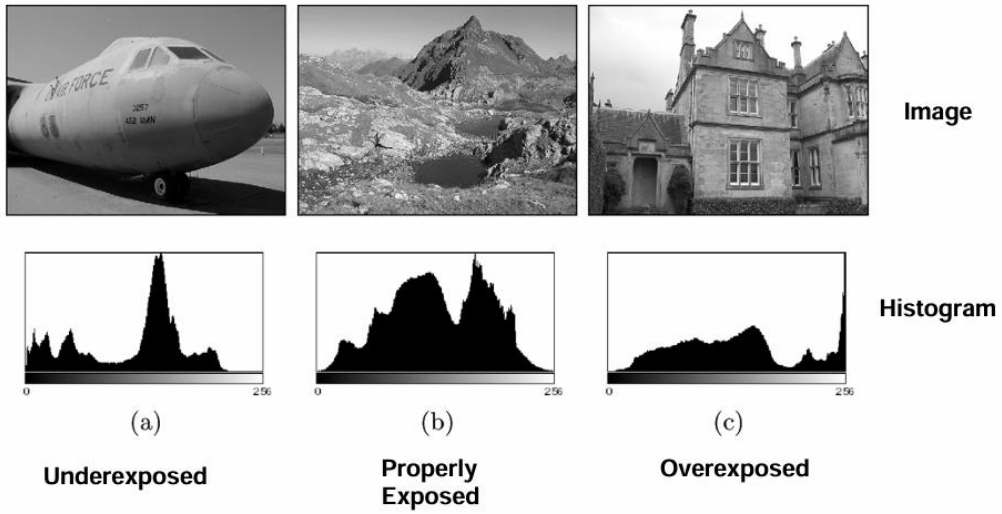
- Different images can have same histogram

Histogram interpretation

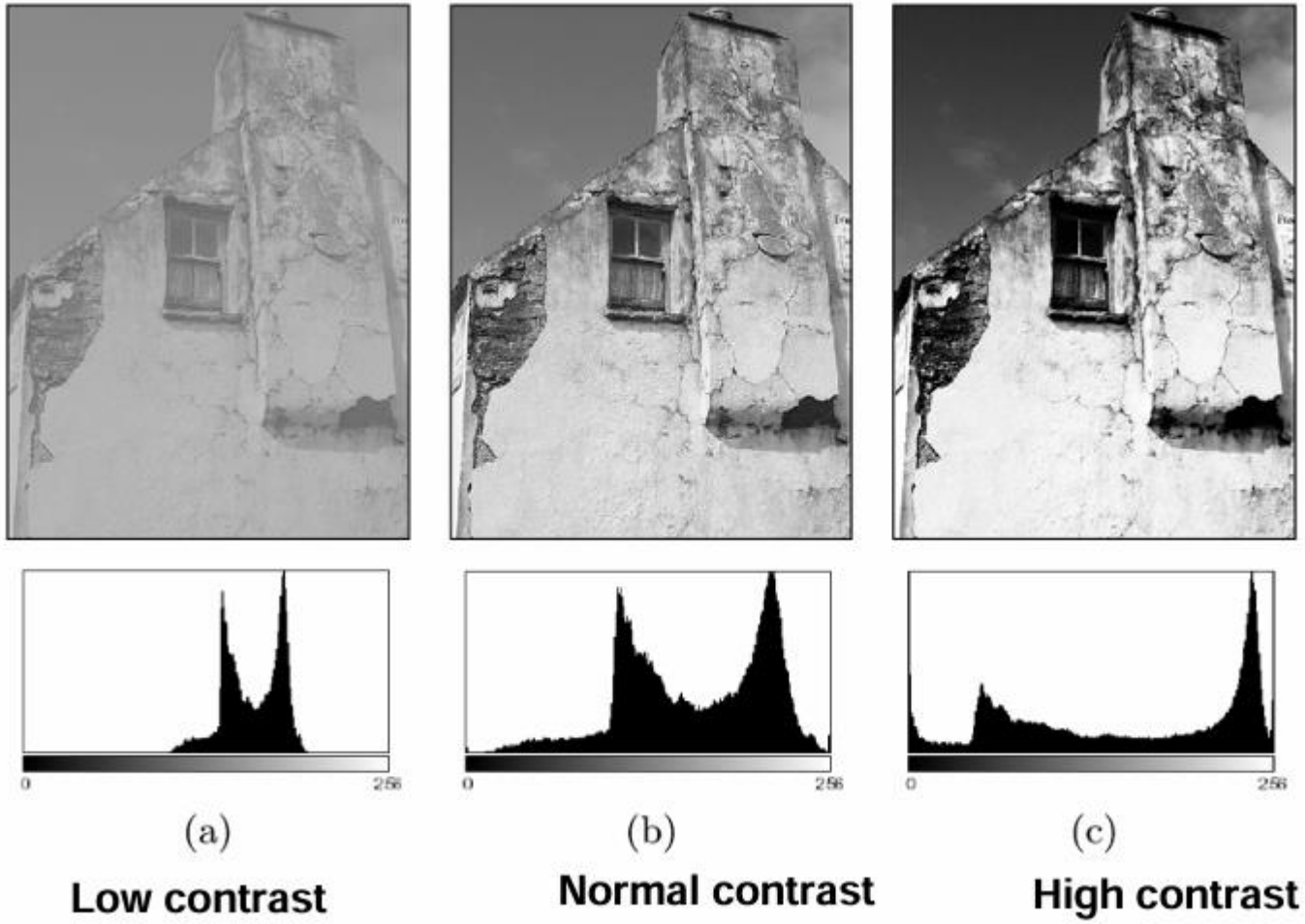
- Log scale to show low values

- Point operations alter the histogran
  - Addition
  - Multiplication
  - Exp, log,
  - …



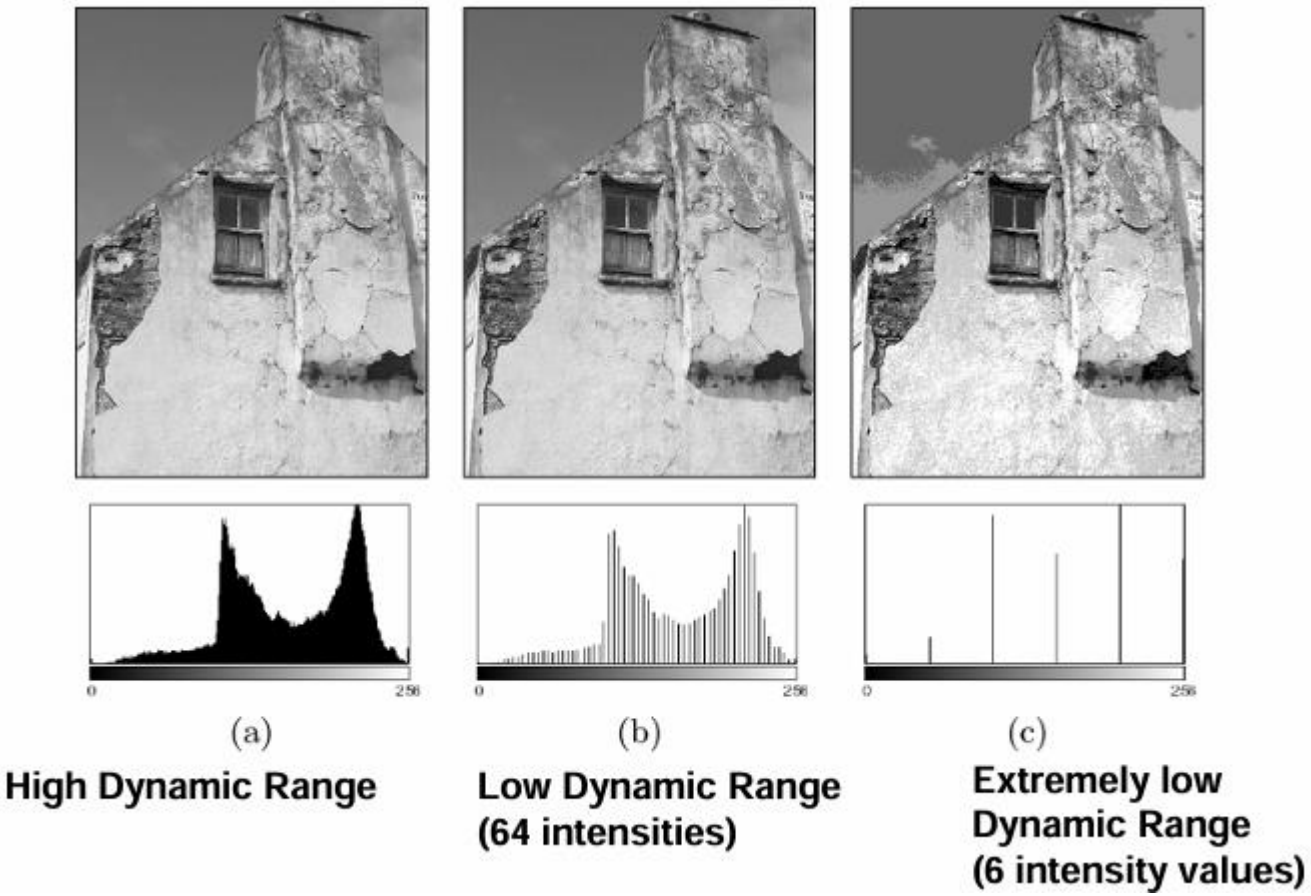


Histogram interpretation - contrast



Histogram interpretation – dynamic range

- Capture at high dynamics range- HDR (12-14 bits) and requantize (8 bits)
- HDR captures very bright and very dark areas within the same image
- Capture multiple exposures and combine

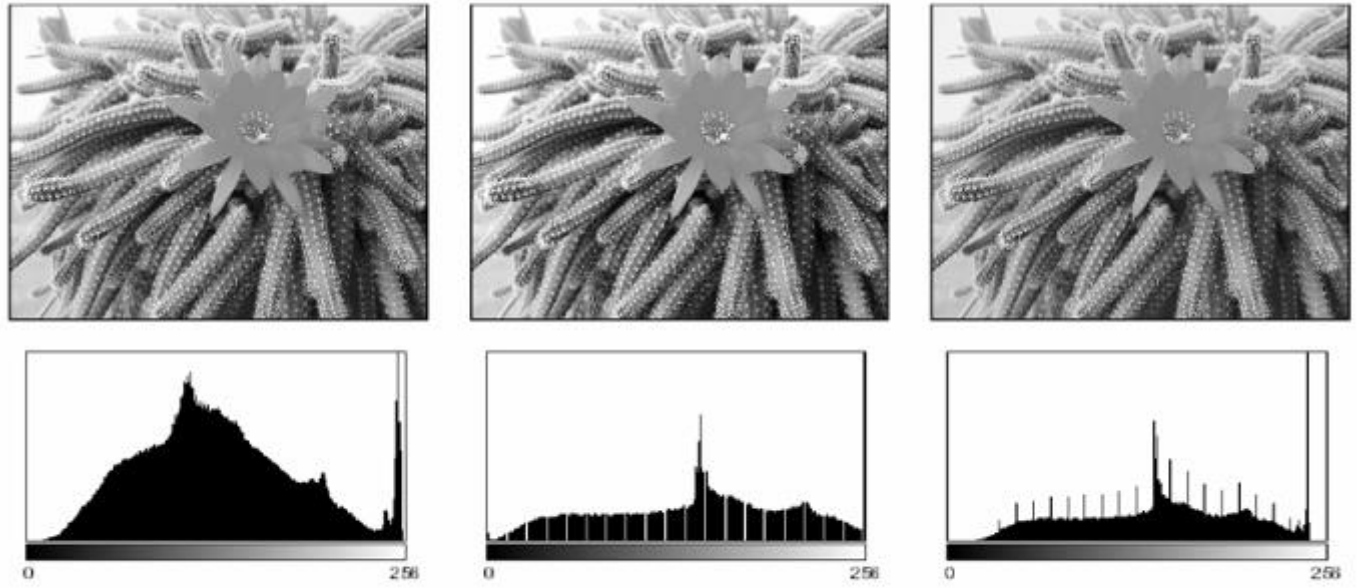


Histogram interpretation – saturation and manipulation

(a) Over-exposure, saturation of processing, …

(b) Quantization, compression, …

(c) Posterization



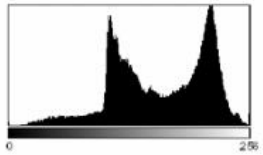
(a) Original (b) Gaps (c) Spikes

Histrogram interpretation - compression

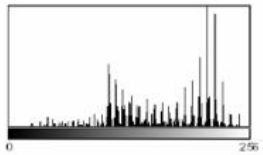
- GIF

Original Image

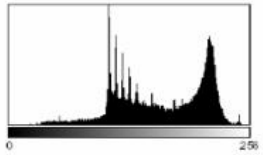




(a) Original Histogram



(b) Histogram after GIF conversion



(c) Fix? Scaling image by 50% and Interpolating values recreates some lost colors

But GIF artifacts still visible

- JPEG



(a)

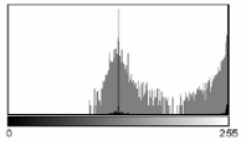


(b)

Original histogram has only 2 intensities (gray and white)



(c)



(d)

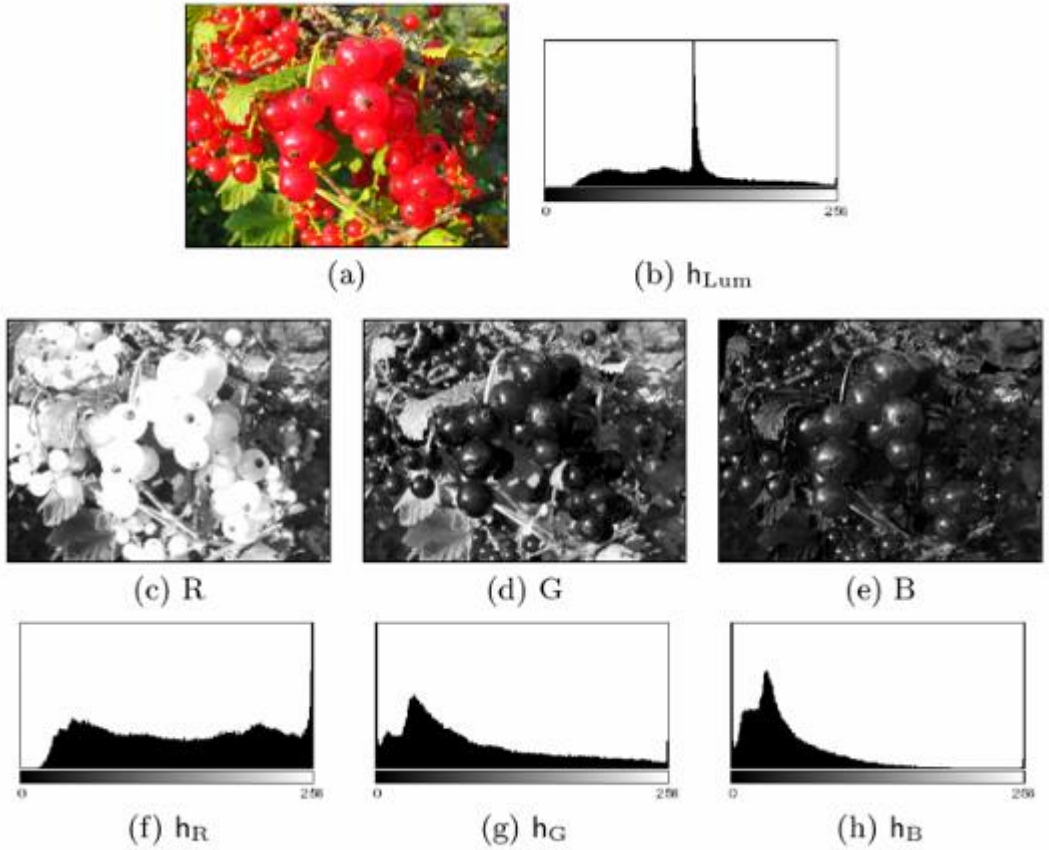
JPEG image appears dirty, fuzzy and blurred

Its Histogram contains gray values not in original

Color histogram

- Images with totally different RGB colors can ha ve same R, G and B histograms

- Hence, shows no information about color distribution



Cumulative histogram

- Analogous to Cumulative Density Function (CDF)

Definition:

$$H(i) = \sum_{j=0}^{i} h(j) \quad \text{for} \quad 0 \le i < K$$

Recursive definition

$$H(i) = \begin{cases} h(0) & \text{for } i = 0 \\ H(i-1) + h(i) & \text{for } 0 < i < K \end{cases}$$

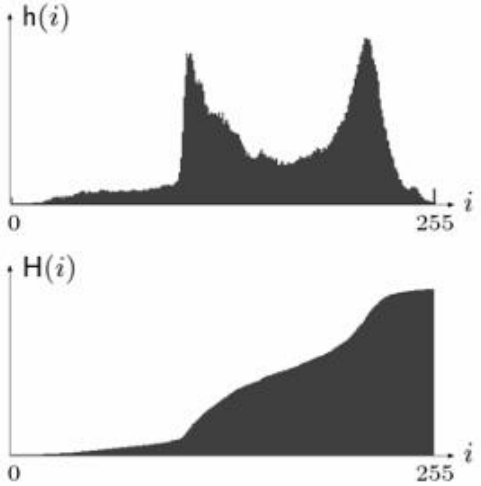
Monotonically increasing

$$H(K-1) = \sum_{j=0}^{K-1} h(j) = M \cdot N$$

Last entry of Cum. histogram

Total number of pixels in image





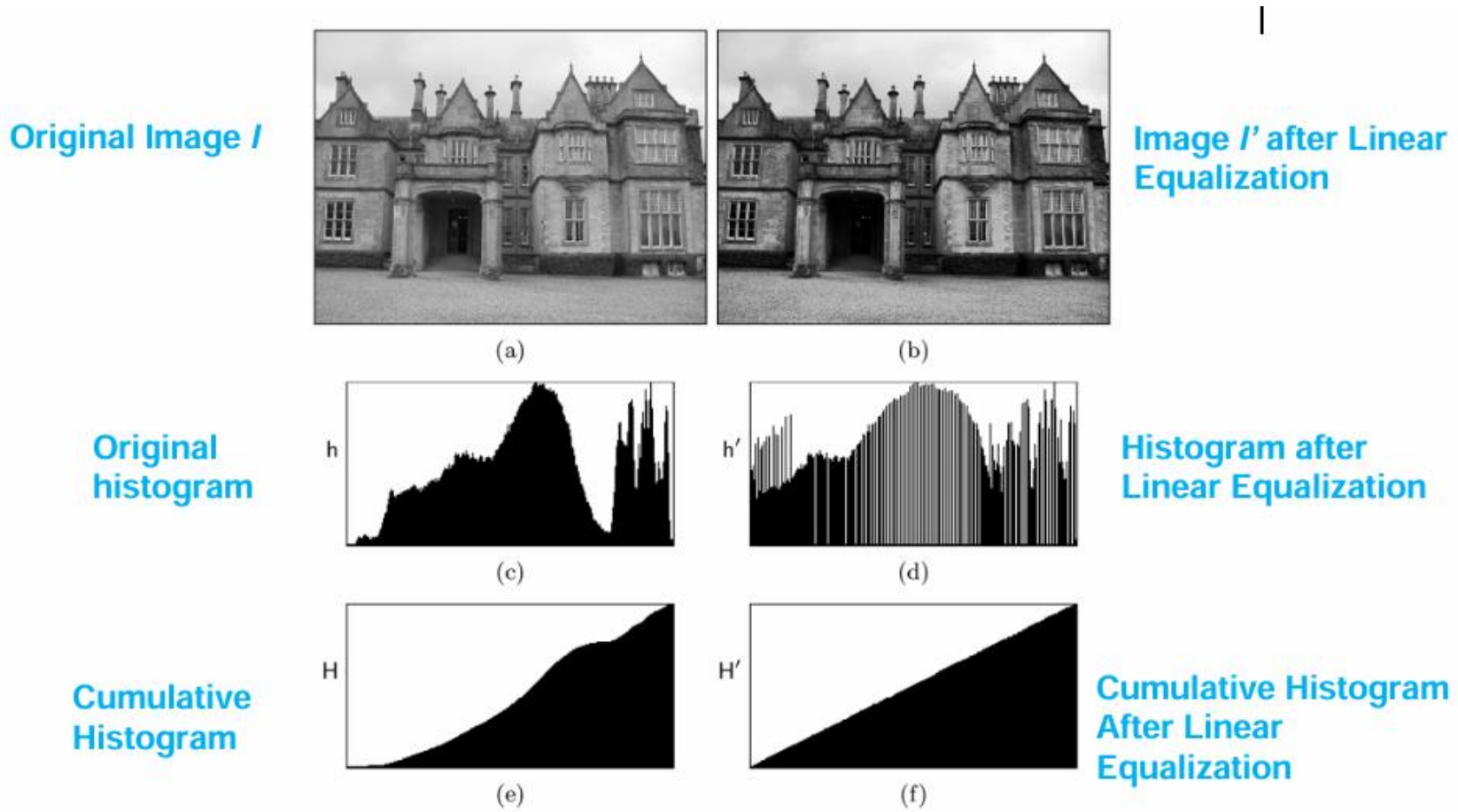
Histogram Equalization – contrast enhancement

Each pixel is processed individually

The cumulative histogram is used as transfer function

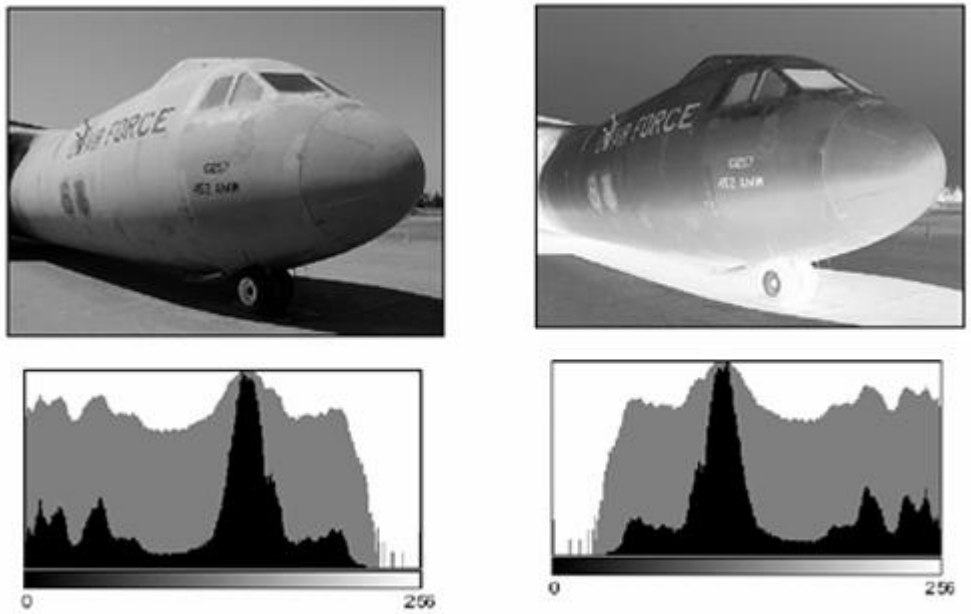
$$s = H(r)$$

where H is the cumulative histogram

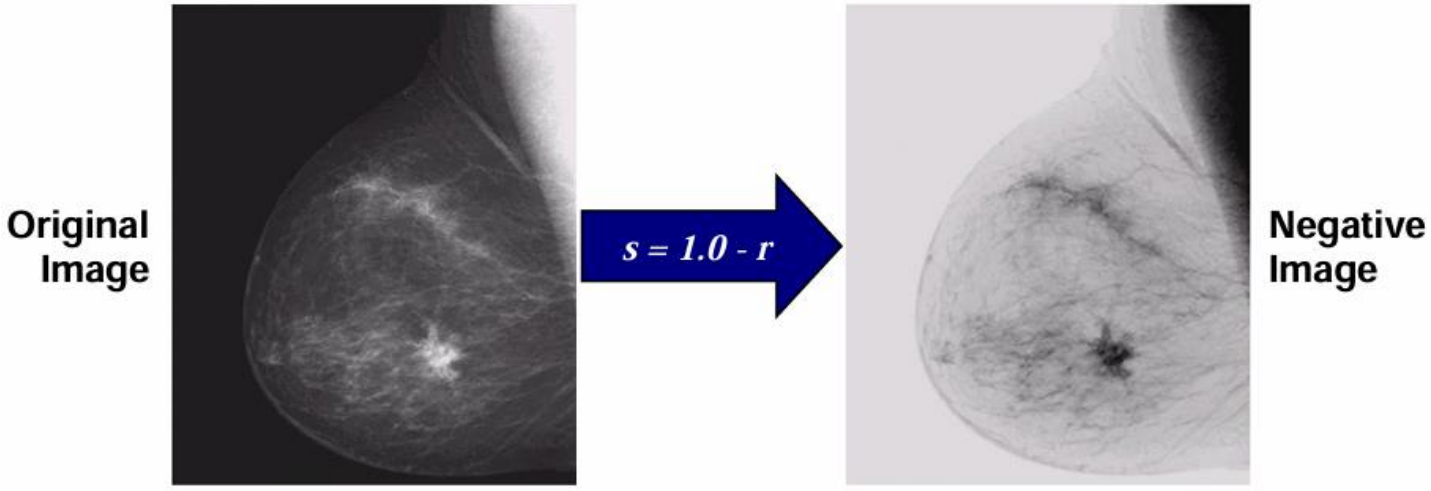


Invert image

$$f_{\text{invert}}(a) = -a + a_{max} = a_{max} - a$$



Original Inverted



Thresholding

- Converts into binary image if $a_0 = 0$ and $a_1 = 1$

$$f_{\text{threshold}}(a) = \begin{cases} a_0 & \text{for } a < a_{\text{th}} \\ a_1 & \text{for } a \geq a_{\text{th}} \end{cases}$$

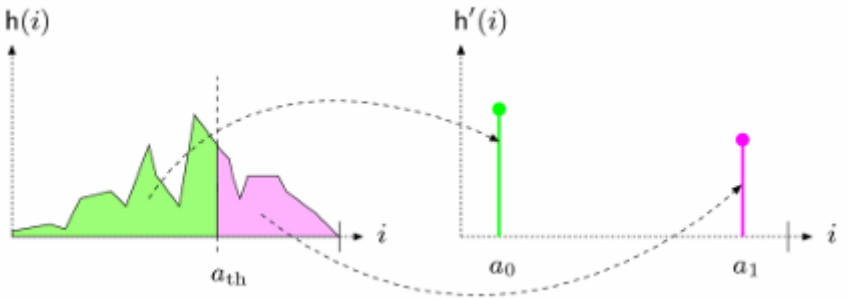


Original Image



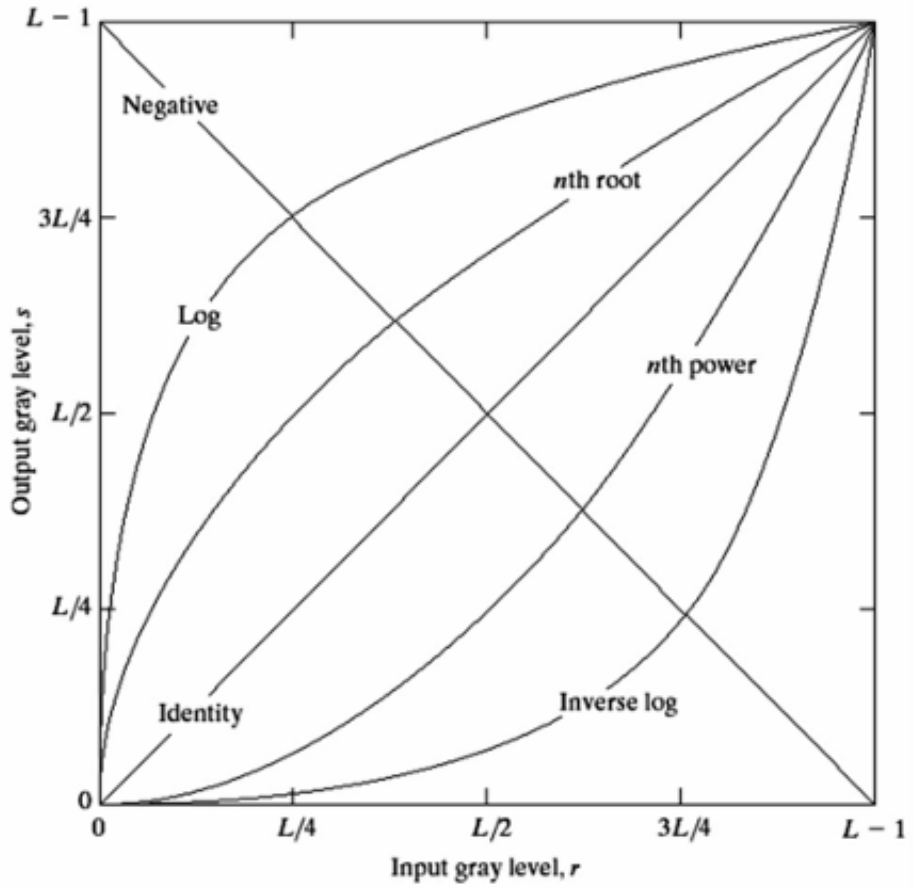
Thresholded Image

- Splits histogram into two halves

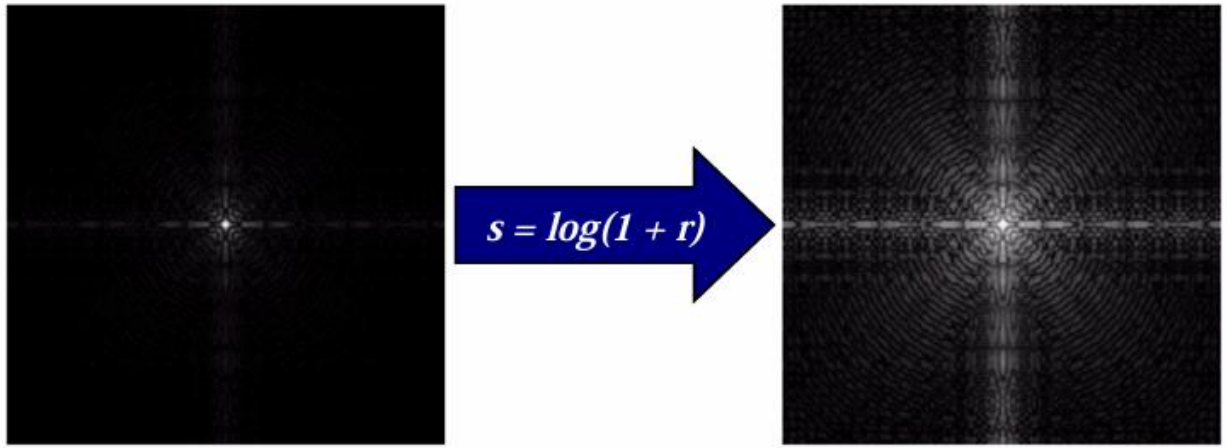


Gray level transformations - log

- $s = c \cdot \log(1 + r)$

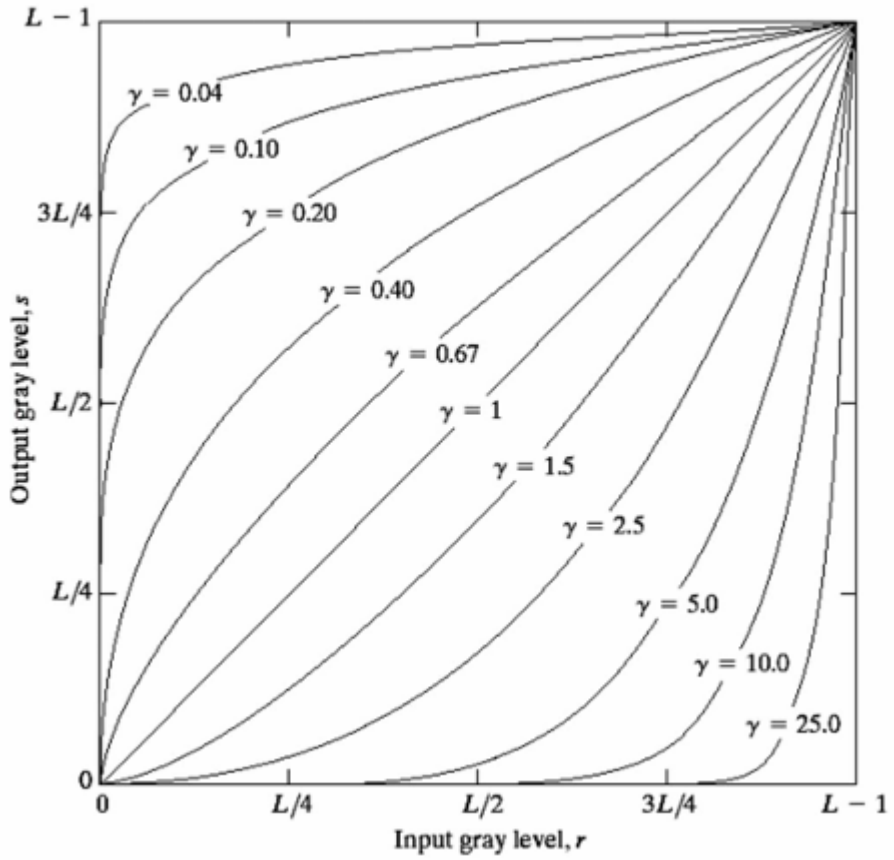


- For example, log of DFT shows more detail

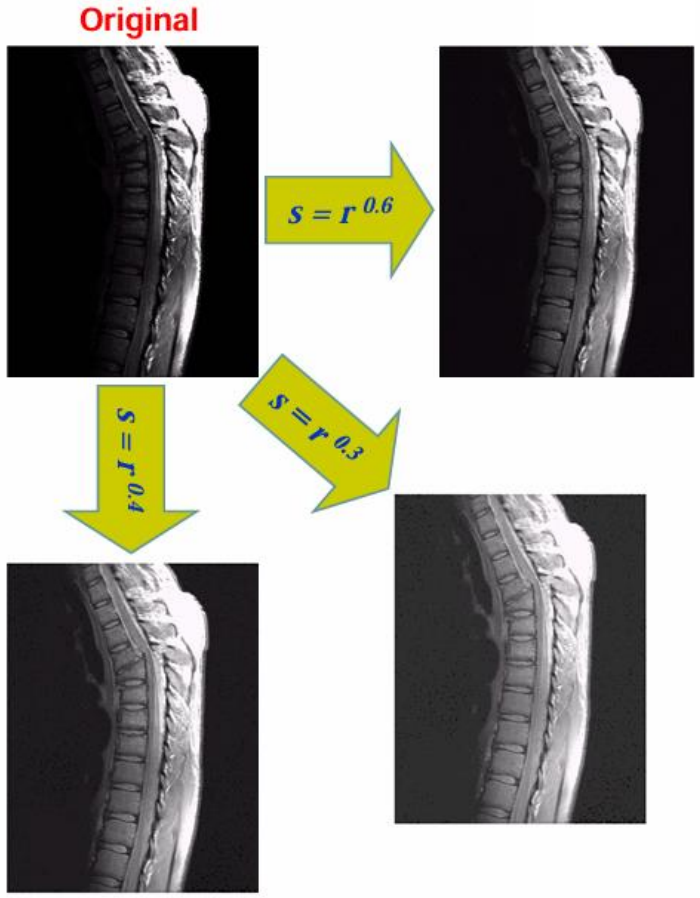


Gray level transformations - power

- $s = c \cdot r^{\gamma}$



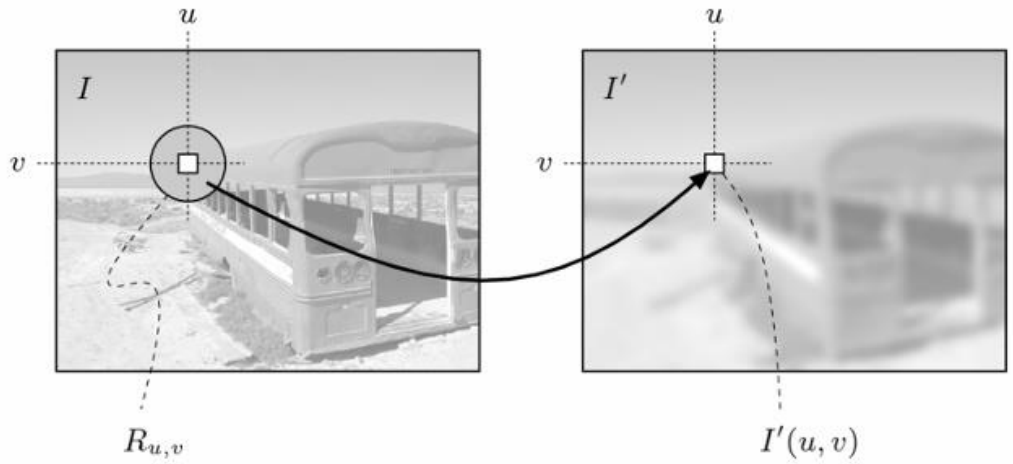
- Magnetic Resonance Image (MRI) of a fractured human spine



Filtering revisited – smoothing by averaging

Many filter parameters to choose:

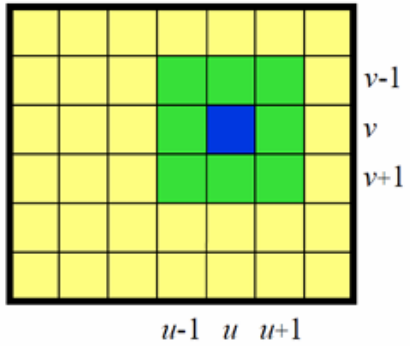
- Size
- Shape (often square)
- Weights
- Function (often linear)



$$I'(u,v) \leftarrow \frac{p_0 + p_1 + p_2 + p_3 + p_4 + p_5 + p_6 + p_7 + p_8}{9}$$

$$I'(u,v) \leftarrow \tfrac{1}{9} \cdot [\, I(u-1,v-1) + I(u,v-1) + I(u+1,v-1) + I(u-1,v) + I(u,v) + I(u+1,v) + I(u-1,v+1) + I(u,v+1) + I(u+1,v+1) \,]$$

$$I'(u,v) \leftarrow \frac{1}{9} \cdot \sum_{j=-1}^{1} \sum_{i=-1}^{1} I(u+i, v+j)$$



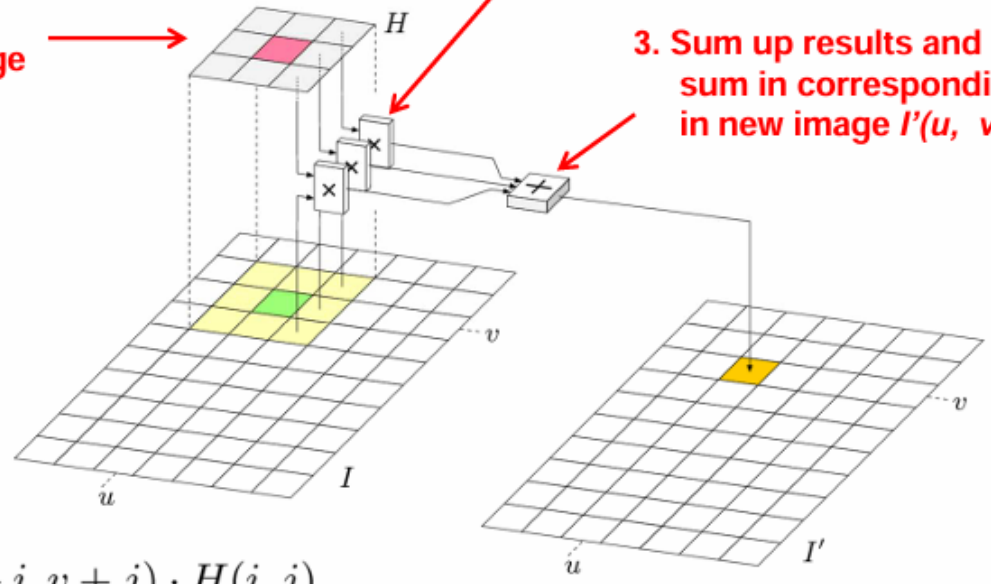
Convolution revisited

For each image position *I(u,v)*:

1. Move filter matrix *H* over image such that *H(0,0)* coincides with current image position *(u,v)*

2. Multiply all filter coefficients *H(i,j)* with corresponding pixel *I(u + i, v + j)*

3. Sum up results and store sum in corresponding position in new image *I'(u, v)*



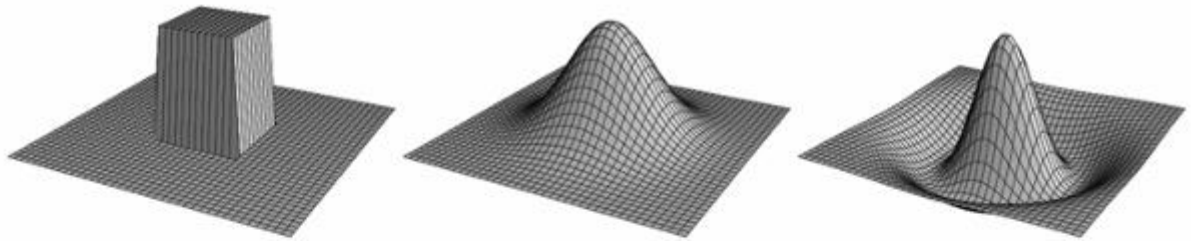
Stated formally:

$$I'(u,v) \leftarrow \sum_{(i,j)\in R_H} I(u+i, v+j) \cdot H(i,j)$$

R_H is set of all pixels Covered by filter. For 3x3 filter, this is:

$$I'(u,v) \leftarrow \sum_{i=-1}^{i=1} \sum_{j=-1}^{j=1} I(u+i, v+j) \cdot H(i,j)$$

Filtering revisited



0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

(a)

Box

0	1	2	1	0
1	3	5	3	1
2	5	9	5	2
1	3	5	3	1
0	1	2	1	0

(b)

Gaussian

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

(c)

Laplacian

smoothing

differencing

Noise

- Type of noise determines best way of removing it



Original



Salt & peppar

Black and white pixels randomly scattered over the image



Gaussian

Normally distributed noise added to image

$I + N$



Speckle

Normally distributed noise multiplied with image

$I(1 + N)$

Courtesy Allasdair McAndrews

Linear noise reduction

- Smearing but no clean removal



Salt & peppar

Black and white pixels randomly scattered over the image



(a) 3×3 averaging

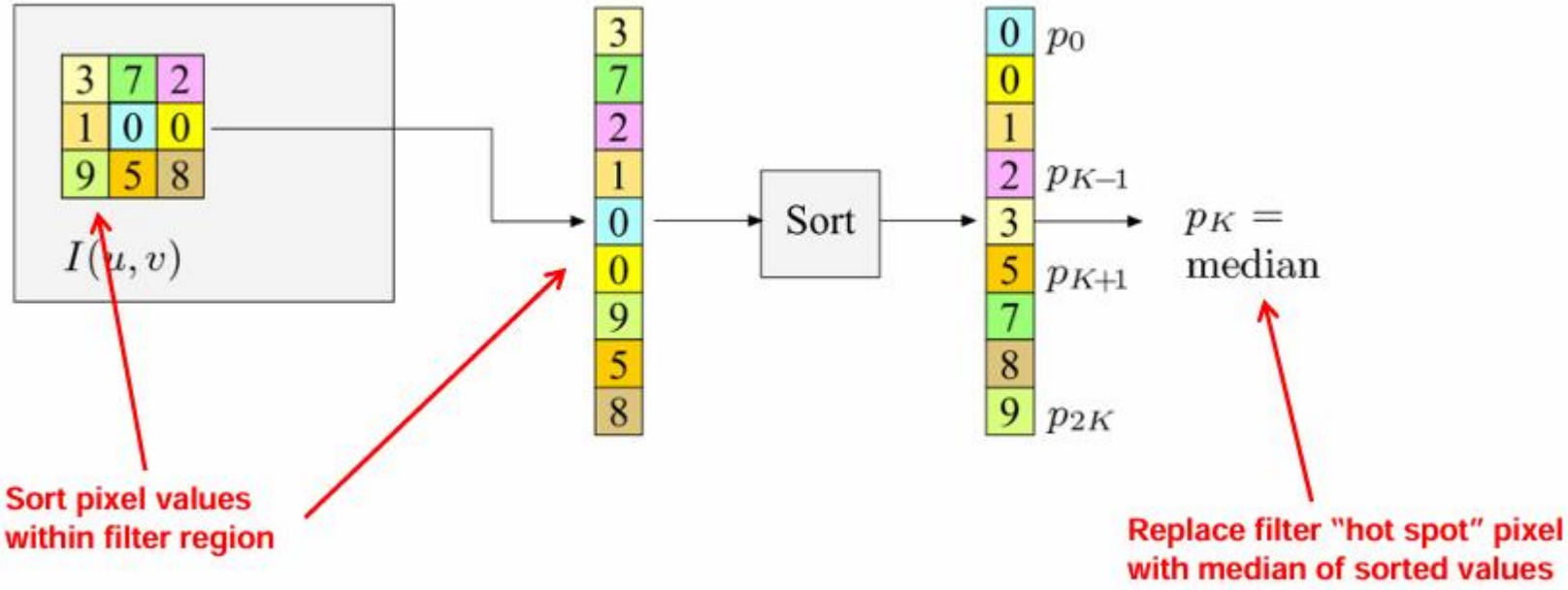


(b) 7×7 averaging

Non-linear filters – median filter

- Better at removing noise while keeping structure

$$I'(u,v) \leftarrow \text{median}\{I(u+i, v+j) \mid (i,j) \in R\}$$



Median filter effects



(a)

Original Image with Salt-and-pepper noise

(b)

Linear filter removes some of the noise, but not completely. Smears noise

(c)

Median filter salt-and-pepper noise and keeps image structures largely intact. But also creates small spots of flat intensity, that affect sharpness

Edge detection - revisited

Image is 2D discrete function

Image derivatives in horizontal and vertical directions

$$\frac{\partial I}{\partial u}(u,v) \quad \text{and} \quad \frac{\partial I}{\partial v}(u,v)$$

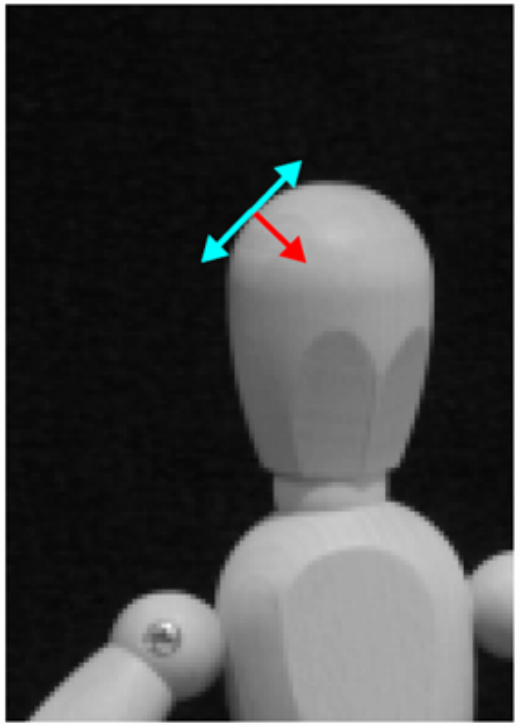
Image gradient at location (u,v)

$$\nabla I(u,v) = \begin{bmatrix} \frac{\partial I}{\partial u}(u,v) \\ \frac{\partial I}{\partial v}(u,v) \end{bmatrix}$$

Gradient magnitude

$$|\nabla I|(u,v) = \sqrt{\left(\frac{\partial I}{\partial u}(u,v)\right)^2 + \left(\frac{\partial I}{\partial v}(u,v)\right)^2}$$

Magnitude is invariant under image rotation, used in edge detection



Derivative approximations - finite differences

Forward difference (right slope)

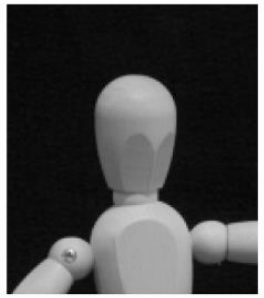
$$\Delta_+ f(x) = f(x+1) - f(x) \qquad \Delta_+ f = f * H = f * [0 \quad -1 \quad -1]$$

Backward difference (left slope)

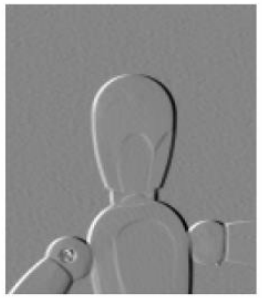
$$\Delta_- f(x) = f(x) - f(x-1)$$

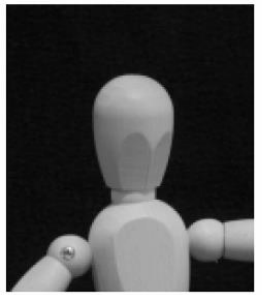
Central Difference (average slope)

$$\Delta f(x) = \frac{1}{2}(f(x+1) - f(x-1)) \qquad \Delta_+ f = f * H = f * [-0.5 \quad 0 \quad 0.5]$$

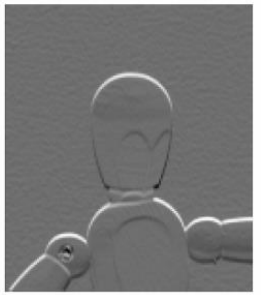


$* [-0.5 \quad 0 \quad 0.5] =$

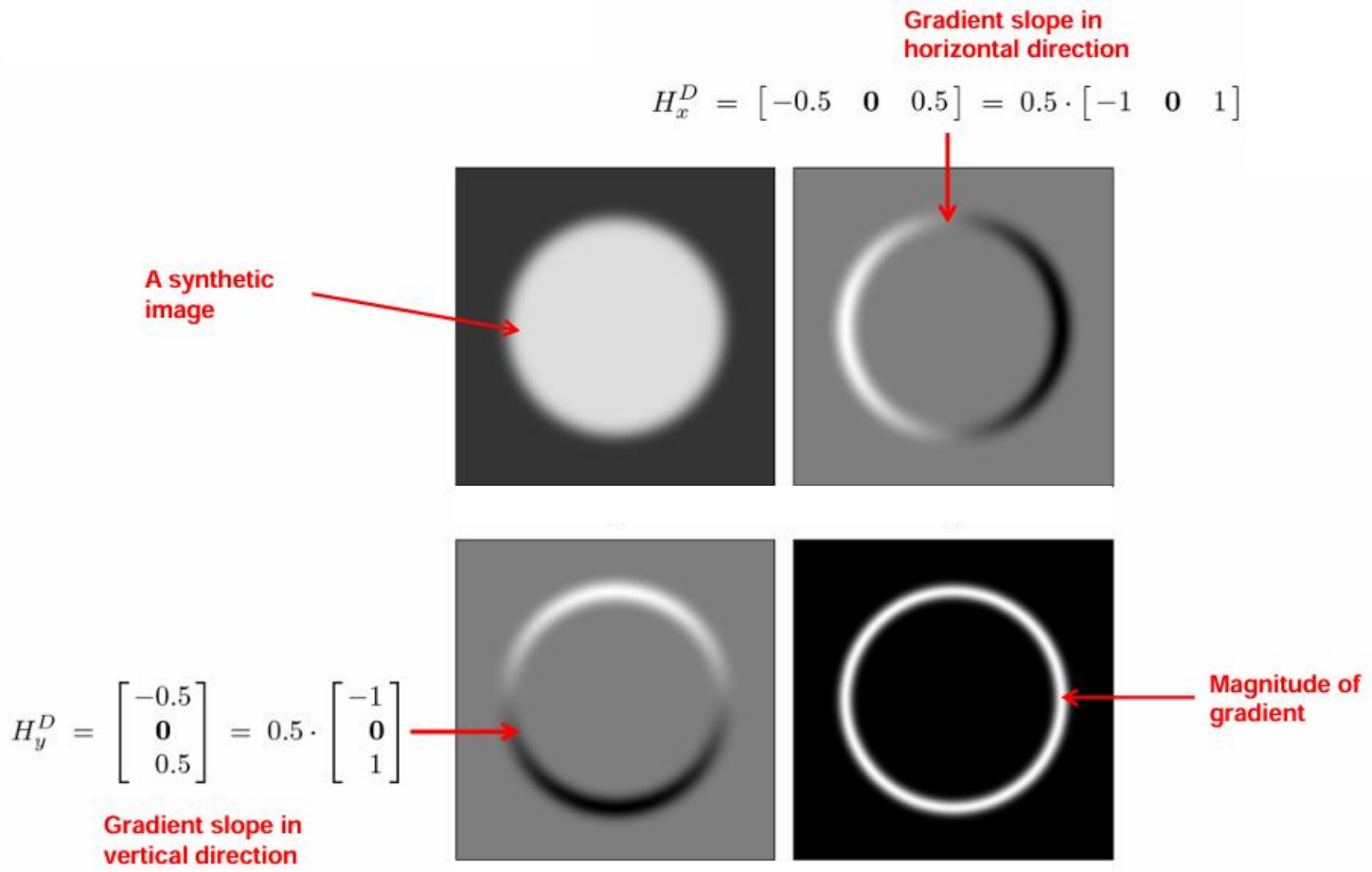




$* \begin{bmatrix} -0.5 \\ 0 \\ 0.5 \end{bmatrix} =$



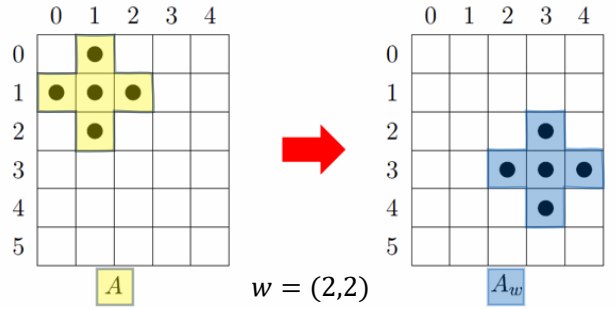
Derivative approximations



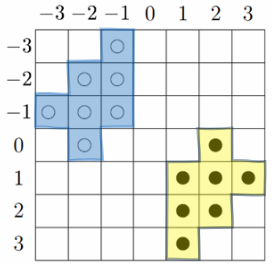
Morphological filters

- Operates on binary images (only black and white pixels) and alters local structures of the image

- Basic operations
  - **Translation**
    A is a set of binary pixels
    $w = (x, y)$ is a particular pixel position
    A_w is set A translated in direction $w = (x, y)$
    $$A_w = \{(a, b) + (x, y) : (a, b) \in A\}$$



  - **Reflection**
    A is a set of binary pixels
    $w = (x, y)$ is a particular pixel position
    A_w is set A reflected by eg.
    $$A_w = \{(-x, -y) : (x, y) \in A\}$$



  - tion

Morphological filters cont.

- Dilation - expands connected sets of 1s
- Erosion - shrinks connected sets of 1s
- Closing - dilation followed by erosion
- Opening - erosion followed by dilation











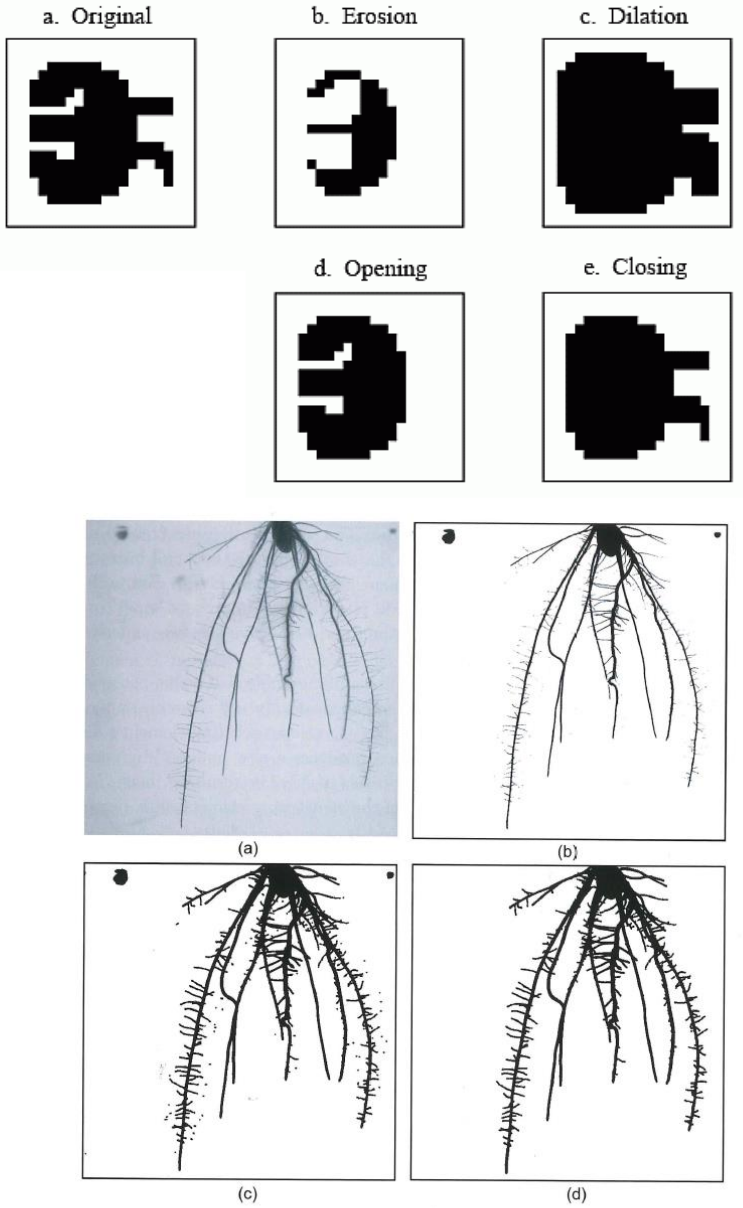
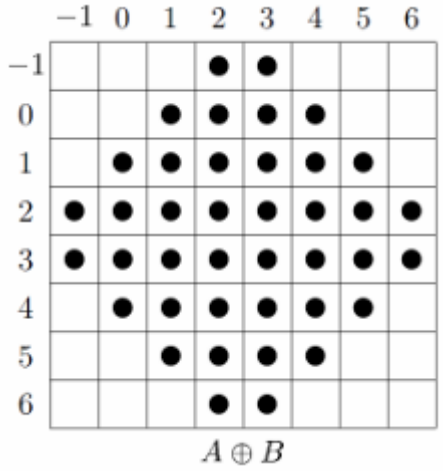


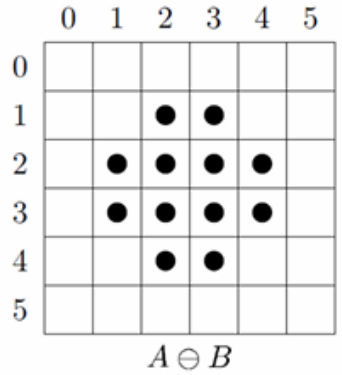
Figure 9.11. Morphological operators. The image in (a) is segmented in (b). (c) Dilation and erosion are used to close holes. (d) The largest connected component is selected. (Data courtesy of R. Peikert, ETH Zürich.)

Morphological filters cont.

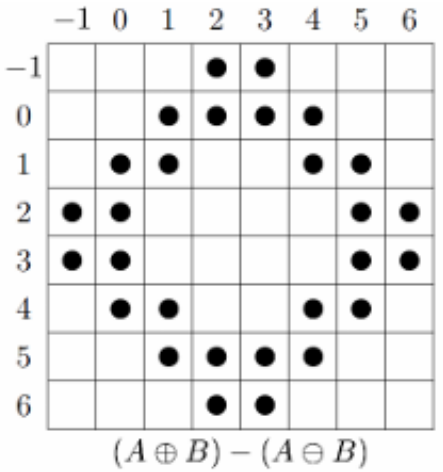
- Boundary detection
- Dilation of Image A – Erosion of Image A



Dilation



Erosion



Dilation - Erosion

- Morphological gradient

Morphological filters examples

- Original
  A
- Internal boundary
  A – Erosion of A
- External boundary
  Dilation of A – A
- Morphological gradient
  Dilation of A – Erosion of A



Original



Internal boundary



External boundary



Morphological gradient

Grayscale morphology



